

This is the report for our project called “Watering indicator for houseplants”. The members of our team are Kristjan Saar, Kevin Noormets and Emil Henri Rooste.

Business understanding

Identifying business goals

Background

The business (Click & Grow) has developed a system that uses the decreasing weight of a potted plant system to deduce soil moisture levels. They have already implemented a linear regression solution to find when the soil goes from phase one drying to phase two drying at which point they can notify the plant owner that the plant should be watered.

Phase 1 drying is fast, primarily surface evaporation of water and Phase 2 drying is slower, indicating the plant is approaching the optimal re-watering point. The goal of this project is to improve the precision of this detection method, as the current linear regression approach is "quite far from precise." The idea is currently waiting for patent approval using the imprecise method, but may change if the new algorithm is much better than the current one.

Business goals

The overarching organizational goal is to optimize the houseplant watering schedule to enhance the performance and commercial value of the product system.

1. **Immediate Technical Goal:** Develop a new predictive algorithm (superior to the current basic linear regression model) that can detect the onset of Phase 2 Drying faster and more accurately than the current implementation. This transition point signals the precise moment when the plant requires re-watering.
2. **Strategic Business Goal:** Deliver an algorithm of such high precision and robustness that it is deemed a viable, superior alternative to the company's current core logic. The optimal outcome is an algorithm that can replace the existing, imprecise patented method, thereby maximizing the reliability and long-term market success of the manufactured product.

Business success criteria

The data-mining outcome must demonstrate a superior technical performance compared to the current system's basic linear regression model:

1. **Precision Improvement:** The new algorithm's statistical accuracy in detecting the Phase 1/Phase 2 transition point must be superior. This can be measured through data analysis and comparison to the original model's results.

2. **Robustness and Reliability:** The model must successfully identify the change point with a higher than 90% accuracy rate and consistent results across multiple different plant types.

The fulfillment of these criteria will be judged by the company's experts.

Assessing your situation

Inventory of resources

This section lists all available resources.

- **People:** We have three people working on this project and the client to ask questions from.
- **Data:** Two years of houseplant weight data, tracking the water loss across eight houseplants.
- **Hardware/Software:** University standard issue laptops, Python environments and relevant machine learning/statistics libraries.

Requirements, assumptions, and constraints

Requirements

1. The solution must detect the phase change point where the weight loss rate decreases significantly.

Assumptions

1. The change in the slope of the weight curve is the reliable physical indicator for the Phase 1 to Phase 2 drying transition.
2. The provided dataset has correct data with clear changes in plant weight over time.

Constraints

1. No supervision from the business, so the project must be completed independently.
2. No new data to test on and no baseline data for verifying our results.

Risks and contingencies

- **Risk: Data Quality/Noise.** The raw weight data contains too much noise, making the subtle change point (Phase 1 to Phase 2 transition) difficult to distinguish from normal measurement fluctuations.

- **Contingency Plan:** Implement robust data smoothing, filtering, and pre-processing techniques to enhance the signal and clarify the slope change before applying the detection model.
- **Risk: Lack of Validation.** Inability to confirm the biological or engineering accuracy of the detected change point without domain expert input due to the project being frozen.
- **Contingency Plan:** Rely entirely on clear statistical metrics (e.g., R^2 , residual analysis, RMSE) and strong visual evidence (clear data segmentation on plots) to provide robust statistical justification for the chosen re-watering threshold.

Terminology

Phase 1 Drying: The initial period of rapid weight loss in the potted system, primarily due to surface evaporation of readily available soil moisture. Characterized by a steep slope on the weight-loss curve.

Phase 2 Drying: The subsequent period of slower, moisture-limited weight loss. Detecting the start of this phase is critical, as it signals the optimal re-watering window.

Costs and benefits

Costs: This project does not involve any direct financial costs for its members. However we account time spent and used computational power as costs associated with this project.

Benefits: Creating a solution superior to an existing one, made by Click & Grow, will allow the company to provide better products to their customers, thus increasing their profits.

Defining data-mining goals

Data-mining goals: The data mining goals for this project are to understand and organize the raw data for it to be used further as training data for training machine learning models that will provide better accuracy in detecting onset of Phase 2 Drying compared to the current implemented solution.

Data-mining success criteria: The data-mining process will be considered successful once the raw data is organized in a meaningful way and the algorithm for detecting Phase 2 Drying is more accurate than the current solution.

Data Understanding

Gathering data

Data requirements

Since the data we are using was provided to us by the company Click & Grow, that data is the only data necessary to address our data mining goals. This data is time series data spanning across 2 years, gathered from numerous different houseplants and soils. Since the goal is to indicate when a plant should be watered based on its weight alone, then the only field necessary is the weight of the plant at a specific moment in time. The data would ideally be in a CSV-format, which is convenient to use for a pandas DataFrame. It would be ideal for there to be data of plants, which were not watered often enough, to check if our model would predict earlier, that the plant should be watered.

Data availability

Since the data was provided to us by a company, and the people at Click & Grow themselves have used this dataset for this exact task, then the data required for this task most certainly exists. We have also confirmed that this data exists and confirmed that we can use it, by importing it into our project and visualizing it. However, since the data for optimally testing our models is unavailable, we are going to try our best to test our model on the available data.

Selection Criteria

Our data was directly provided by Laurits Puust from Click & Grow. The fields most relevant for our cause are all of them, since this dataset only has the weight values and entry times. There are many anomalous data points in this data, which we are going to exclude. For example moments where there were no plants on the scales or electricity blackouts.

Describing data

The dataset consists of 46647 rows of data, at first taken at 30 minute intervals, and later at 5 minute intervals. The 9 fields consist of the time of the moment the data was gathered, and the weight results (in grams) of 8 scales, which had 8 plants on them. We are going to be using all of these fields (field1 to field8) for our project. Since the data gathered with 5

minute intervals (rows 22674 to 46647) is more accurate, then we are going to focus more on those data points, rather than the ones collected at 30 minute intervals.

The coordinator at Click & Grow also noted that not all of the data provided is of the same quality, since at times over-watering, electrical blackouts and plants leaning on each other occurred. Another important thing to note is that we have no way of telling if a plant was watered too early, too late, or just at the perfect time. This complicates the evaluation of the accuracy of our model.

Exploring data

The created_at column of the data ranges from the 6th of December 2023 until the 19th of June 2025. However more than half of the data points come from the range of 26th March 2025 until 19th of June 2025, since the interval between entries was changed. The lowest weight value reported was -689.3 grams and the highest was 2047.1 grams. Each field has quite a few instances of negative data points, which are quite illogical in the context of weights of plants, so we are going to have to deal with those values. There are many values near 0 grams as well, since the plants were moved and scales were tested sometimes. These weights are again not very useful. The means of the weights range from 231.7 to 579.6 from scale to scale, which is quite logical for the weight of a houseplant. We also noticed quite a few instances of the weight values suddenly going up or down for one entry, only to return to the previous value for the next entry. We have also developed algorithms to find when a plant was watered, and to identify a cycle from the point the plant was watered until the next watering.

Verifying data quality

We pointed out most of the data quality issues in the “Exploring data” section of this report. While these data quality issues are quite annoying, they are not detrimental to this project, since we can either edit the wrong values, or just use only the fully valid data, which we have quite a lot of. Then again, since we can never be sure from the data, if the plants were watered at correct intervals, it is quite hard to find data to **test** our models on.

Plan

Task 1: Initial research, coordination with Click & Grow (6 hours)

This is initially getting to know the data, understanding what it means, understanding what our goals are, and how this goal has previously been achieved. Also this accounts for writing emails to Click & Grow. Emil - 3h, Kristjan and Kevin - 1.5h.

Task 2: Data preparation (50 hours)

This task is crucial in our case, since the data is quite confusing and it has many faults. For this we are going to be using algorithms, which detect correct watering cycles in the data, which detect illogical events in the data. It is also part of this task to deal with the fact that the dataset is in 2 parts: 30-minute intervals and 5-minute intervals. From that data we must also find usable intervals with as few anomalies as possible in order to accurately find the phase change point. Kristjan 20h, Emil 15h, Kevin 15h

Task 3: Modeling (15 hours)

This means actually using the prepared data in a meaningful way to complete the task. Currently we plan to try using change point detection with linear regression, and residual analysis. We are also going to experiment with using the differences in weights. Emil 6h, Kevin 7h, Kristjan 2h

Task 4: Evaluation (15 hours)

Assessing whether our model performed well, tweaking it, testing it. Evaluating if we have reached our business goals. Spotting issues we have overlooked. Emil 6h, Kevin 7h, Kristjan 2h.

Task 5: Poster (6 hours)

Visualizing our results using graphs and including them on a poster to communicate our efforts, challenges, data and results in a simple and understandable way, while keeping enough complexity to deliver the technical understanding of the tasks to the reader. Kristjan - 4h, Kevin 1h, Emil 1h.

Task 6: Team coordination - organizing meetings, version control, communication (2-3 hours)

We account for about 1 hour per person for these things.